KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF ORTHOPAEDIC TECHNOLOGY

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN ORTHOPAEDIC TECHNOLOGY

PAPER: BIOMECHANICS

DATE: 5TH FEBRUARY 2025 **TIME: 3 HOURS (9:00AM – 12:00 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the EXAMINATION **NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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SECTION ONE: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. What factor can you consider when selecting an assistive device for a 60-year-old woman with a history of rheumatoid arthritis?
   A. Cost of the device
   B. Cosmesis (appearance) of the device
   C. Patient's weight and activity level
   D. Severity of joint damage and functional limitations
2. A young athlete with a transtibial amputation struggles with knee flexion during prosthetic gait training. You suspect a socket fit issue. What assessment would you consider to determine the problem?
   A. Patient's pain level during gait training
   B. Radiographs of the residual limb
   C. Gait analysis video focusing on knee joint motion
   D. Measurement of the patient's thigh circumference
3. What factors would you consider in choosing a prosthetic foot for a diabetic amputee?
   A. Patient's age and weight
   B. Patient's desired activity level and risk of falls
   C. Color and style preference of the foot
   D. Availability of different foot models in the clinic
4. What would you recommend to trans-femoral amputee with phantom pain?
   A. Prescribe a strong pain medication
   B. Fit a temporary prosthesis immediately
   C. Recommend mirror therapy and desensitization techniques
   D. Refer the patient for psychological counseling only
5. What is the intervention of a patient with a below-knee prosthesis walking with a circumduction gait pattern?
   A. Educate the patient on proper gait mechanics
   B. Adjust the prosthetic components to improve alignment
   C. Recommend gait training with a physical therapist
   D. Immediately replace the prosthetic foot with a different design
6. What pathological gait pattern is characterized by the swinging of the body over the limb to allow clearance, often seen in patients with a prosthesis or limb discrepancy?
   A. Scoliosis
   B. Tight limb abductors
   C. Adopted style on walking
   D. Contralateral shortening
7. Which components should you consider to provide optimal protection and support when designing a modified shoe for a patient with diabetic foot ulcers?
   A. Sole, heel, upper lining, and reinforcement
   B. Vamp, upper, sole, and toe box
   C. Quarters, heel, vamp, and shank
   D. Toe box, upper quarters, and lining

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8. What does the term interface mean in prosthetics?
   A. Joint facets
   B. Muscle insertion
   C. Socket wall – stump surface contact
   D. Socket contours
9. What is the purpose of applying a rigid dressing in the early stages of prosthetic management for an amputee patient?
   A. Stop bleeding
   B. To add weight on residual limb
   C. To protect infections
   D. Control edema
10. Which type of prosthesis is characterized by an internal supportive structure that can be easily adjusted or modified?
   A. Crustacean
   B. Exoskeletal
   C. Pegleg
   D. Modular
11. What is the origin for the hamstrings muscles, crucial for performing knee flexion and hip extension?
   A. Pubic tubercle
   B. Ischial tuberosity
   C. Posterior gluteal line
   D. Greater sciatic notch
12. What biomechanical issue result in delayed knee flexion in a transtibial prosthesis user, potentially leading to gait inefficiencies?
   A. Flexed socket
   B. Foot – ankle assembly placed posteriorl
   C. Posterior tilted socket
   D. Toe – break posteriorly
13. What technique should a transfemoral amputee use to safely descend stairs while marking time to ensure stability and reduce the risk of falls?
   A. Shift weight to the prosthesis and lower the sound limb
   B.]Keep weight to the sound limb and lower the prosthetic limb
   C. Keep weight supported by crutches
   D. Use long leg supports
14. What gait deviation in a transfemoral amputee is typically caused by inadequate knee flexion due to plantar-flexion of the prosthetic foot?
   A. Dorsiflexion
   B. Inversion
   C. Eversion
   D. Plantar-flexion

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15. What is the accumulation of fluid in the soft tissues, often resulting in swelling and discomfort, indicative of an underlying pathological condition?
    A. Sinus
    B. Edema
    C. Boil
    D. Viscous
16. Which position is effective for observing the mid-stance phase of gait to assess alignment and stability?
    A. Best observed sideways
    B. Best observed from rear
    C. Best on mid-swing
    D. Aerial view
17. How is compensation of missing limb achieved in to restore function and alleviate discomfort in orthopaedic technology?
    A. Prosthetics
    B. Orthotics
    C. By extension
    D. Braces
18. What type of joint axis is depicted by transtibial prosthesis attached with side joints and thigh corset?
    A. Multi-axial
    B. Di-centric
    C. Polycentric
    D. Mono-centric
19. What can cause prosthetic foot rotation at heel strike, leading to instability and gait deviations?
    A. Soft heel cushion
    B. Patient habit
    C. Socket angulation
    D. Hard heel cushion
20. What is the cause of early knee flexion in a transtibial prosthesis user which can lead to instability and inefficient gait?
    A. Soft heel cushion
    B. SACH foot keel placed in front
    C. Harder heel cushion
    D. Knee extensors contracture
21. What is the purpose of an extended Thomas heel in orthopaedic footwear?
    A. To treat calcaneal spur
    B. To replace lateral arch support
    C. To replace medial arch support
    D. To treat a painful sole

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22. What is the typical external rotation angle of the ankle joint due to the natural torsion of the tibia?
    A. 5-7 degrees
    B. 10-15 degrees
    C. 20-30 degrees
    D. 10-40 degrees
23. Which of these is a non-functional knee joint used in transfemoral prosthesis to provide stability and control?
    A. Mechanical
    B. Pneumatic
    C. Hydraulic
    D. Polycentric
24. What is a common reason for the amputation of a grossly deformed but functionally good leg?
    A. Weight bearing
    B. Cosmetic
    C. Shortening
    D. Vaulting
25. Which model of shoe last closely resembles the natural shape of the human foot, providing a better fit?
    A. Custom shoe last
    B. An out-flare shoe last
    C. A regular shoe last
    D. An in-flare shoe last
26. What is a reverse Thomas heel used for in orthopaedic footwear?
    A. Medial arch support
    B. Transverse arch-support
    C. A lateral longitudinal drop of an arch-support
    D. Subtalar – eversion
27. Where is a rocker bar used to facilitate walking, especially in patients with restricted forefoot motion?
    A. In Prosthesis
    B. In Plaster cast
    C. In Polio boot
    D. In Compensated orthotics
28. Which spinal condition is characterized by an exaggerated inward curve of the lumbar spine?
    A. Lordosis
    B. Kyphosis
    C. Scoliosis
    D. Spondylolisthesis

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29. What is the area between the ball of the foot and the anterior border of the heel?
    A. Shank piece
    B. Shank area
    C. Ball of the foot
    D. Vamp
30. What are purpose of surgical foot wear?
    A. Footwear to minimize pressure on sensitive deformed structures
    B. Function and cosmetic
    C. Protection and healing device
    D. Dance
31. What is the anatomical terms for mid sagittal plane?
    A. The sagittal
    B. The median
    C. The frontal
    D. The transverse
32. What causes the sinusoidal curve trail generated by the center of gravity during normal human locomotion?
    A. Knee flexion
    B. Pelvic dip
    C. Torso
    D. Lordosis
33. How is the level of the iliac spine maintained during walking?
    A. Keeping leg straight
    B. Replacing a lost part of the body
    C. Compensation in orthotics
    D. The rate of normal walking
34. What limits joint movement, potentially leading to decreased range of motion?
    A. Bone shape
    B. Position
    C. Flesh contact
    D. Type of the stump
35. Which muscles are primarily responsible for extending the knee?
    a) Hamstrings
    b) Calf
    c) Extensor medially
    d) Quadriceps
36. What is the deformity of the hip joint, where the femur is angled inward?
    A. Coxa-valga
    B. Coxa-vara
    C. Genu-valga
    D. Genu-vara

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37. What is the component whose proximal end is attached to the socket and distal end to the foot-ankle assembly in early prosthetic management?
    A. Modular
    B. Pylon
    C. Pipe
    D. Prosthesis
38. What is the process of arrangement of prosthetic and orthotic components to ensure proper alignment and function?
    A. Progression
    B. Alignment
    C. Organization
    D. Weight line
39. During which phase of normal human locomotion is the amplitude of the center of gravity pronounced?
    A. Double support phase
    B. Weight acceptance
    C. Single support phase
    D. Push off phase
40. What type of orthosis is used to provide support and immobilization to the spine?
    A. Brace
    B. Belt
    C. Band
    D. Stays

SECTION TWO: SHORT ANSWER QUESTIONS (40 MARKS)

41. Discuss five (5) early intervention following lower limb amputation. **(5 Marks)**
42. Outline three (3) similarities and two (2) differences when the fitting prostheses to transtibial versus transfemoral amputations by. **(5 Marks)**
43. Explain how weakness in the hip lateral rotators can contribute to difficulty in climbing stairs after transfemoral amputation. **(5 Marks)**
44. Explain five (5) movements at the shoulder joint. **(5 Marks)**
45. Explain five (5) characteristics of normal human gait **(5 Marks)**
46. Discuss five (5) factors which would influence your choice of suspension method for transtibial prosthesis. **(5 Marks)**
47. Identify five (5) causes of circumduction in a patient using a knee ankle foot orthosis when walking. (5 Marks) **(5 Marks)**
48. Explain five (5) ways in which the design of a transfemoral socket can influence both prosthetic stability and sensory feedback for the user. **(5 Marks)**

SECTION THREE: LONG ANSWER QUESTIONS (20 MARKS)

49. Susan is a 35-year-old female who contracted poliomyelitis at the age of 4. As a result of the disease, she presents with flaccid paralysis of her right lower limb. She has a noticeable limb shortening of 9 cm, which significantly impacts her mobility and quality of life. She currently uses a walking aid but is seeking an orthotic solution to improve her functional mobility and daily activities. Discuss orthotic intervention that will be suitable in managing Susan's condition. **(20 Marks)**

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